

Alessandro Caula

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Profile

Versatile Software Developer with expertise in C#, Python, and JavaScript/Typescript, and a passion for creating efficient, data-driven solutions for scientific and full-stack projects. Skilled in software engineering, algorithm design, data processing, and visualization, I build robust and user-friendly applications. Committed to continuous improvement, my adaptability, creativity, and clear communication have helped me develop tools that enhance research and support users.

Technical Skills

Programming Languages: C#, Python, JavaScript/TypeScript, Bash, R, HTML, CSS

Frameworks & Full-Stack: .NET, React, Node.js/Express, Tailwind CSS, MongoDB

Machine Learning & Data Science: NumPy, Pandas, Scikit-learn, TensorFlow, Biopython, Jupyter Notebook

Visualization: Matplotlib, Seaborn, Plotly, DevExpress, LightningChart

OS & Tools: Linux, macOS, Windows, Git, VS Code, Google Colab, AWS, Docker,

Experience

Bioinformatics Software Developer, 3Brain AG – Genova, ITALY

May 2022 – Sept 2025

Designed and developed high-performance scientific software for real-time electrophysiological data acquisition, analysis and visualization in *BrainWave*, a C#/.NET application used by neuroscience researchers. Key contributions:

- Built interactive and intuitive GUIs and charts for real-time data exploration and visualization.
- Implemented and integrated statistical and machine learning algorithms into core pipelines.
- Optimized algorithms for signal processing, frequency analysis ($3\times$ faster) and trajectory detection.
- Efficiently processed large datasets (50–80 GB/recording) through scalable pipelines, ensuring high performance.
- Engineered modular, object-oriented components, ensuring scalability and maintainability.
- Worked in a large, fast-moving codebase, collaborating across teams to deliver maintainable, robust software.
- Improved performance and robustness through rigorous debugging and architectural refactoring.

Data Scientist Intern, University of Lisbon – Lisbon, PORTUGAL

Apr 2021 – Feb 2022

Developed and implemented supervised and unsupervised Machine Learning pipelines for omics data analysis using Python, aiming to identify molecular subtypes of Alzheimer's disease ([[GitHub](#)] [[Paper](#)]).

- Preprocessed and cleaned data, performed feature selection and modeling, and evaluated results with key metrics.
- Created visualizations and summaries of model outputs, clearly communicating biological insights to researchers.
- Applied a range of ML methods including Non-negative Matrix Factorization, Support Vector Machines, Naïve Bayes, k-Nearest Neighbor, Decision Trees, Logistic Regression, Random Forests, and XGBoost.

Laboratory technician, IHMA Sarl – Monthey, SWITZERLAND

Feb 2019 – Jul 2019

Education

University of Bologna, M.Sc. in Bioinformatics

Oct 2019 – Feb 2022

Gained a strong foundation in computational biology, algorithms implementation, data science, and machine learning.

- **Key coursework:** Algorithms and Data Structures, Python and Bash programming, Basic and advanced Machine Learning, Systems and in Silico Biology, Big Data Processing Infrastructures, DNA/RNA dynamics

University of Turin, B.Sc. in Biology

Sept 2015 – Oct 2018

Projects & Courses

Some personal projects to deepen my expertise and refine my skills, available on my [GitHub](#):

- **Fullstack Open (University of Helsinki):** Hands-on fullstack development course covering React, Node.js, REST APIs, TypeScript, testing, and styling. *Strengthened skills in modular architecture and building end-to-end applications.*

Languages

Italian - Native

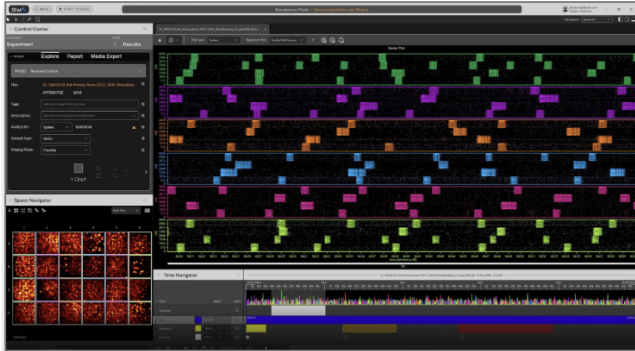
English - Fluent (professional proficiency)

French - Beginner

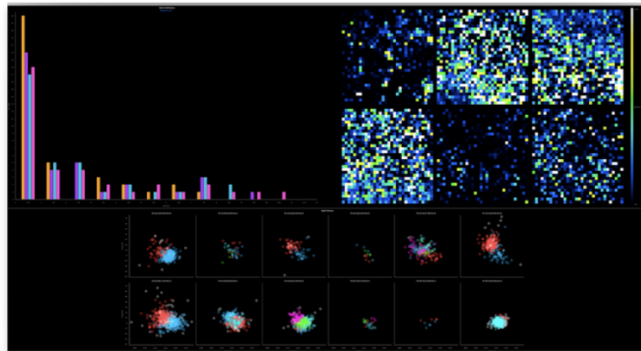
Project Highlights

BrainWave - 3Brain AG

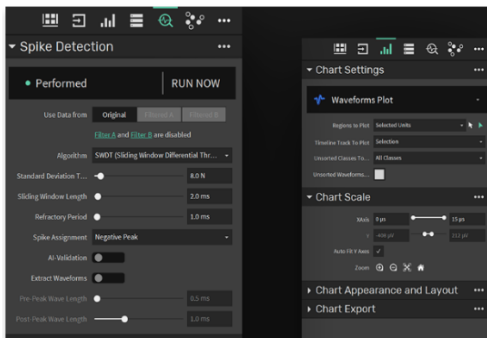
High-performance C#/.NET software for real-time electrophysiology data analysis, and visualization.



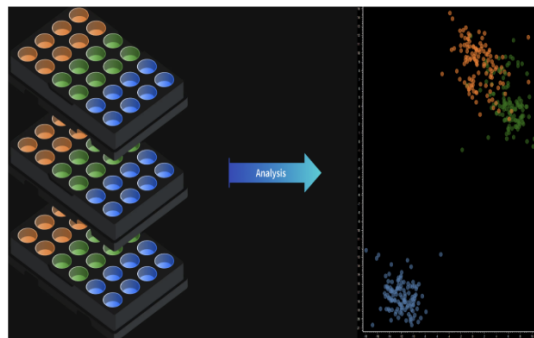
Modular, high-performance software with interactive GUI components.



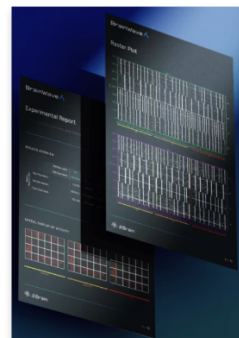
Interactive and dynamic visualizations of electrophysiological analysis.



Fully customizable and intuitive graphical interface.



ML and clustering on multi-channel recordings.



Automated Report.

Python Structural Modeling Pipeline

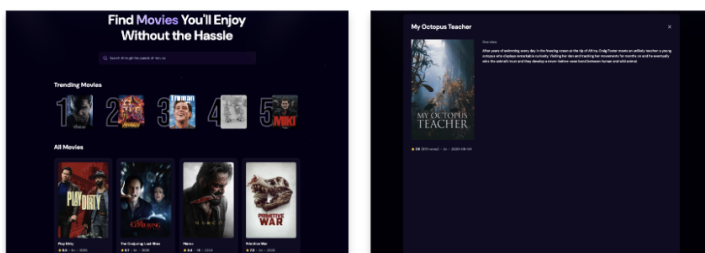
Python pipeline for automatic identification and modeling of missing internal loops in protein structures:

- Generated multiple loop conformations with the CCD algorithm, and evaluated for steric clashes, Ramachandran outliers, and anchor RMSD.
- Designed modular, extendable pipeline for structural bioinformatics applications.

Web Development

Practiced front-end and full-stack web development using **JavaScript**, **TypeScript**, **React** and modern libraries, applying best practices for modular and maintainable code.

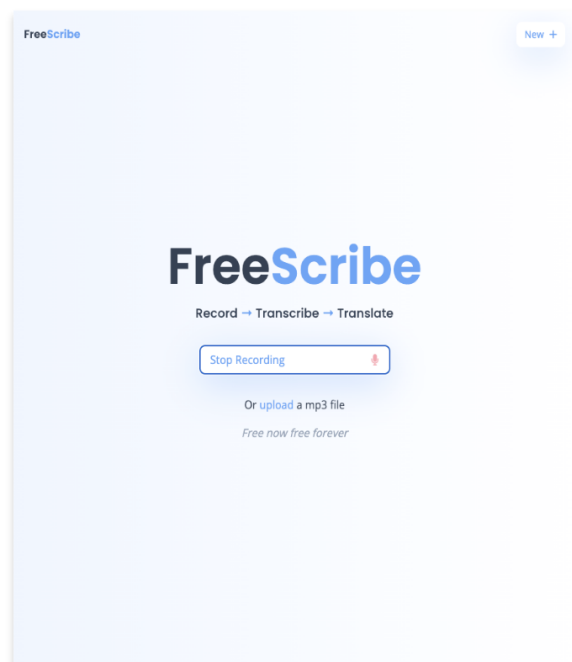
- Strengthened skills in modular frontend architecture and user-friendly interface design.
- Worked with APIs, backend communication, and real-time processing (including Web Workers for AI tasks).



React app for browsing and discovering movies via external API.



Modern landing page built with React, Tailwind, and GSAP.



React app for audio-to-text translation using OpenAI.